

EduGenie – Google Gemini Powered Learning Assistant

Project Description

EduGenie is an AI-powered educational learning assistant designed to provide personalized academic support for students and self-learners. The application enables users to ask educational questions, receive simplified concept explanations, generate multiple-choice quizzes, summarize lengthy study materials, and obtain personalized learning roadmaps.

Built using FastAPI, Jinja2, Google Gemini 1.5 Flash, LaMini-Flan-T5, SQLite, and SQLAlchemy, the system offers a secure and responsive web application with JWT authentication and learning history management. EduGenie helps learners improve their understanding, prepare for examinations, and organize their learning through intelligent AI assistance.

Scenarios

Scenario 1

A college student preparing for semester examinations enters a difficult programming question into EduGenie. The system generates a detailed explanation, provides a simplified concept using LaMini-Flan-T5, and recommends additional study topics to improve understanding.

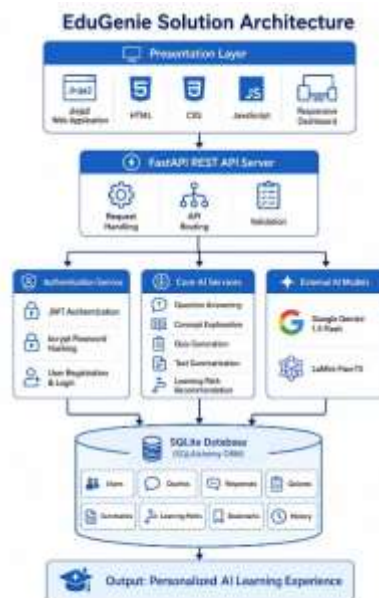
Scenario 2

A competitive exam aspirant uploads a lengthy study topic and requests a summary. EduGenie generates a concise summary, creates multiple-choice questions for practice, and recommends a personalized learning roadmap.

Scenario 3

A self-learner exploring Artificial Intelligence asks questions about Machine Learning concepts. EduGenie answers the questions using Google Gemini, explains difficult concepts in simple language, stores the learning history, and allows bookmarking important responses.

Technical Architecture :



Description:

EduGenie follows a modular three-tier architecture.

- The **Presentation Layer** consists of a Jinja2-based web interface developed using HTML, CSS, and JavaScript.
- The **Backend Layer** is implemented using FastAPI and manages API routing, authentication, and request processing.
- The **Service Layer** contains reusable modules for AI orchestration, authentication, prompt management, quiz generation, summarization, and learning path generation.
- The **Data Layer** uses SQLite with SQLAlchemy to store users, quizzes, summaries, learning history, bookmarks, and activity logs.

The architecture ensures scalability, modularity, maintainability, and secure AI-powered educational services.

Pre-requisites

- Python Programming Knowledge
- FastAPI Framework
- HTML, CSS and JavaScript
- Jinja2 Template Engine
- SQLite Database
- SQLAlchemy ORM
- Google Gemini API
- Hugging Face Transformers
- REST API Concepts
- Git and GitHub
- Virtual Environment Management

Project Workflow

Milestone 1: Environment Setup and Architecture Design

Activity 1.1: Define Project Requirements

- Identify student learning requirements.
- Define AI-powered educational features.
- Finalize project scope

Activity 1.2: Design System Architecture

- Create architecture diagram.
- Define interaction between frontend, backend, AI services, and database.

Activity 1.3: Set Up Development Environment

- Install Python.
- Create virtual environment.
- Install project dependencies.

Activity 1.4: Create Project Structure

Create folders for:

- app/
- api/
- database/
- models/

- schemas/
- services/
- static/
- templates/

Milestone 2: Backend Development

Activity 2.1: Implement FastAPI Backend

Create API endpoints:

- /
- /register
- /login
- /qa
- /explain
- /quiz
- /summarize
- /roadmap
- /history
- /bookmark

Activity 2.2: Develop AI Question Answering Module

Implement Google Gemini Integration.

Functions:

- answer_question()
- Responsibilities:
- Process educational questions.
 - Generate accurate AI responses.

Activity 2.3: Develop Concept Explanation Module

Implement LaMini-Flan-T5.

Functions:

- explain_concept()
- Responsibilities:
- Simplify complex concepts.
 - Generate beginner-friendly explanations.

Activity 2.4: Develop Quiz & Learning Modules

Functions:

- generate_quiz()
 - summarize_text()
 - generate_learning_path()
- Responsibilities:
- Generate quizzes.
 - Summarize study material.
 - Recommend learning roadmaps.

Milestone 3: Frontend Development Activity

Activity 3.1: Develop Jinja2 Web Interface

Create pages for:

- Landing Page

- Registration
- Login
- Dashboard
- Quiz
- Summary
- Learning Path
- History

Activity 3.2: Add User Interaction Features

Implement:

- Learning History
- Bookmarks
- Quiz Evaluation
- Retry Quiz
- Loading Animations
- Responsive Dashboard

Milestone 4: Data Base and Logging

Activity 4.1: Database Design

Create normalized tables:

- Users
- Queries
- Responses
- Quizzes
- Summaries
- Learning Paths
- Saved Responses
- History
- Activity Logs

Responsibilities:

- Store educational records.
- Maintain user history.
- Store AI-generated content.

Activity 4.2: User Activity Logging

Responsibilities:

- Record login activities.
- Store bookmarks.
- Maintain learning timeline.

Milestone 5: Testing and Validation

Activity 5.1: Unit Testing

Create test files:

- test_backend.py

Validate:

- API functionality
- Database connectivity
- Authentication
- AI integration

- Quiz generation
- Learning Path generation

Activity 5.2: Swagger API Testing

Access Swagger UI:

<http://127.0.0.1:8000/docs>

Test:

- Authentication APIs
- AI APIs
- History APIs
- Bookmark APIs

Milestone 6: Deployment and Documentation

Activity 6.1: Local Deployment

Run application:

```
python -m uvicorn app.main:app --reload
```

Access application:

<http://127.0.0.1:8000/>

Activity 6.2: Version Control

Upload project to GitHub.

Repository includes:

- Source Code
- Documentation
- Images
- Database Schema
- Tests
- README

Exploring the Application Pages

Landing Page

Description

The landing page allows users to:

- Learn about EduGenie features.
- Register or log in.
- Navigate to the dashboard.

Question Answering Section

Description

Allows users to enter educational questions and receive AI-generated answers.

Example:

Question:

What is Machine Learning?

Output:

A detailed explanation generated by Google Gemini.

Concept Explanation Section

Description

Generates simple explanations for difficult academic concepts using LaMini-Flan-T5.

Quiz Generation Section

Description

Generates multiple-choice questions based on the selected topic.

Users can:

- Attempt quizzes.
- View scores.
- Retry quizzes.

Summarization Section

Description

Produces concise summaries from lengthy educational text.

Learning Path Section

Description

Generates personalized learning roadmaps from beginner to advanced levels.

History & Bookmarks Section

Description

Displays:

- Previous AI responses
- Saved bookmarks
- Learning history

Conclusion

EduGenie successfully demonstrates the practical application of Generative AI in education by providing intelligent question answering, concept explanation, quiz generation, text summarization, and personalized learning path recommendations through a unified platform.

Built using **FastAPI**, **Google Gemini 1.5 Flash**, **LaMini-Flan-T5**, **SQLite**, and **SQLAlchemy**, the application follows a modular, secure, and scalable architecture. Features such as JWT authentication, learning history, bookmarks, and a responsive web interface enhance the overall learning experience. Future enhancements include cloud deployment, Retrieval-Augmented Generation (RAG) with PDF uploads, multilingual support, voice-based interaction, mobile application development, and advanced learning analytics, enabling EduGenie to evolve into a comprehensive AI-powered educational platform.